PLANETARY COUNCIL · POLICY BRIEFING · OCTOBER 2026

Can Britain Feed Itself?

Store and grow: a simple plan to keep Britain fed



Everyone eats, every day. Britain grows just enough food in a normal year, keeps no food in reserve, and depends on a long chain of ships, gas, lorries and good weather. For about **£10 a person a year** we could keep a national food cupboard for everyone. For about **£4.50 more** we could look after the soil, water and seed that every future harvest needs.

STORE

A national food cupboard

~£10.33 per person per year

GROW

Healthy soil, water and seed

~£4.46 per person per year of new money

Every number has a small footnote; the sources for each page are listed at its foot and are clickable. Our models, notes and the full evidence file: github.com/planetarycouncil/food-security-climate-regeneration. **Tell us where this breaks.**

Why food comes first

> You can put off almost anything for a while. Not dinner.

Every one of us runs on food. An average woman needs about 2,000 calories a day and an average man about 2,500.[1] We can wait for a new phone, a train, a pay rise. We cannot wait long for food, and neither can our children.

Most of us never think about where the next meal comes from, because it has always been there. The shops are full, the lorries come at night, the shelves are restocked by morning. But shops keep only about ten days of food.[2] Our kitchens hold less. The food system works like a river, not a lake: it is safe only while it keeps flowing.

The good news is that food, unlike electricity, can be **stored**. Dry grain kept cool and clean lasts for years. That is one of the oldest ideas we have. In the Bible's story, Joseph tells Pharaoh to save grain in seven good years to get Egypt through seven bad ones.[3] Finland, Switzerland, Norway, Sweden, Germany, Japan and Singapore still do a version of this today.[4] Britain does not.[5]

"There are only nine meals between mankind and anarchy."

Alfred Henry Lewis, 1896[6]

~10 days

of food in the shops at any time. After that, everything depends on the next delivery.[2]

A VERY OLD IDEA

Store in good years, eat in bad ones. Grain, rice, oats and dried beans keep well. Japan keeps rice for about five years before selling it on.[4]



0

national food stores in Britain, and "no plans to create these" (UK government, March 2026).[5]

So: food is the one thing we cannot go without, and one of the few things we can save up. This briefing asks a simple question: should Britain save some?

1. NHS, *Understanding calories*
2. Full Fact, the UK's food supply
3. Genesis 41 (NIV), Bible Gateway
4. Country details and sources on page 12 (chapter 8)
5. UK Parliament, written answer HL15829 (31 Mar 2026)
6. Quote Investigator, "nine meals from anarchy"

THE SHORT VERSION

The whole story on one page

> Britain grows just enough. Just enough is not safe. Here is what we propose.

In a normal year, British farms grow about as many calories as the country eats: 101% of what we need, a spare of 31 calories per person per day, less than half a slice of bread.[1] That thin margin depends on things we do not control: fertiliser made from imported gas, animal feed from soya grown across the ocean, lorries that arrive just in time, and weather that is getting less reliable.

When one link breaks, the shelves can empty in days. When several break together, a bad season can become a bad year. The government says it holds no national food stocks.[2] So we propose two things, both cheap, both needed.

1 · STORE: A NATIONAL FOOD CUPBOARD

~£10.33 per person a year.[3] Enough wheat to bake bread for everyone for months. A three-day pack of food that needs no cooking, for every person in the country. Baby milk and special foods. Seed and fertiliser for next year's harvest. Rice, oil and sugar.

It protects us for weeks and months: blockades, power cuts, cyber attacks, failed imports.

2 · GROW: SOIL, WATER AND SEED

~£4.46 per person a year of new money, plus today's £4.87bn of farm support aimed at keeping us fed.[3,4] Test and heal the soil, store winter rain, breed our own seed, reuse nutrients, grow more of our own protein.

It protects us for years and decades: less dependence on imports, more ability to cope with drought.

~£14.79

per person a year for both: **about 4p a day**.[3]

A PLAN WRITTEN IN ADVANCE

A clear, step-by-step plan for what happens in a crisis: who gets what, fair shares for all, decided calmly now, not in a panic later.

WHAT TO DO NOW

1. Pass a law that sets a minimum food reserve and clear rules for releasing it.
2. Check how the official statisticians would count the reserve in public debt, before buying anything.
3. Test it: mills, local delivery, and a ration designed by dietitians.
4. Start testing soil and storing rainwater on farms.

How to read this. Each page tells one part of the story and ends with a bold line that leads to the next. Numbers are in the boxes; sources are at the foot of each page.

1. Defra, *Nutritional supply of agriculture in the UK 2025* (15 Oct 2026); bread comparison ours (a medium slice is roughly 80–100 calories)
2. UK Parliament, written answer HL15829 (31 Mar 2026)
3. Our cost model, `reserve_basket.py`
4. Defra, *Agriculture in the UK 2025*, ch. 10–11

What safety is worth

> When safety works, nothing happens. That is exactly why it is easy to skip paying for it.

A fire extinguisher that is never used looks like wasted money, until the day it is needed. A food reserve is the same. If it is never used, nothing happens, and nothing happening looks like nothing bought. The cost comes now; the disaster it prevents might come later and land on someone else. Mark Carney, the former Governor of the Bank of England, called this the "tragedy of the horizon".[1]

So we do not promise huge benefits. We ask a more honest question: **how small a benefit would be enough** to pay for it? The answers below are those break-even points, not forecasts. Each one alone would be enough, so they should never be added up. First number: store only. Second: store and grow.

PROTECTING THE FOOD INDUSTRY

0.44% · 0.63%

of what the food industry adds to the economy each year. It is worth £162bn and employs 4.1 million people.[2]

LIVES

~290 · ~410

deaths prevented a year, on average, would pay for it, using the value the government already uses for road safety.[3]

A STEADY ECONOMY

~8 · ~5 years

of the plan cost the same as one week in which the country produces 10% less (about £5.6bn).[4]

PEACE OF MIND

Making people's lives feel even slightly better would pay for it, by an amount far too small to measure (Treasury wellbeing method).[5]

CALM STREETS

Not counted in the numbers. In the 2000 fuel blockade, supermarkets warned they were days from running out.[6] Rising prices and unfair treatment, more than empty fields, are what make people angry.

COMPARED WITH DEFENCE

~1p : £1

Britain spends about £890 per person a year on defence.[7] The food cupboard costs about £10: roughly 1p on food safety for every £1 on military safety.

So: the plan pays for itself even if its benefits are tiny. Now the real question: how likely is it that we will need it?

1. M. Carney, *Breaking the tragedy of the horizon*, Bank of England (2015)
2. Defra, *Food statistics in your pocket* (food industry value £162.3bn, 2024; 4.1 m jobs); £718m ÷ £162bn (our arithmetic)
3. DfT, RAS60: value of preventing road deaths (£1.8m in 2016 prices, raised to ~£2.5m: our update)
4. Our model, `benefits()` in `reserve_basket.py`
5. HM Treasury, *Wellbeing guidance for appraisal*: break-even is 0.0008–0.0011 points on a 0–10 life-satisfaction scale
6. CNN transcript, UK fuel protests (13 Sep 2000)
7. House of Commons Library, CBP-8175, UK defence spending (2026) (~£62bn a year)

CHAPTER 1

Just enough

On paper, Britain can feed itself. In practice, the margin is wafer-thin.

Forty years ago, Britain grew more than 80% of the food it ate. Today it is about 60%.[1] We grow plenty of milk, lamb, oats and barley. But we grow only 83% of our wheat, just over half of our fresh vegetables and about one piece of fresh fruit in seven.[1]

Look closer at our animals. Pigs and chickens eat about 1.1 million tonnes of protein a year from soya shipped in from abroad.[2] Cows and sheep use most of our farmland.[3] If the soya ships stopped, the spare calories would shrink, as the government's own statisticians point out.[4]

And the people who grow our food are getting older. Four in ten farmers in England are over 65; only one in twenty is under 35.[5] When food prices jumped in 2023, they rose by almost a fifth in a single year.[6]

60%

of the food we eat is grown in the UK (by value, 2025). It was over 80% in the 1980s.[1]

101%

of the calories we need are grown here in a normal year: a spare of just 31 calories a day each.[4]

STRONG AND WEAK SPOTS

We grow **83%** of our wheat, **56%** of our fresh vegetables and **14%** of our fresh fruit.[1]



READ WITH CARE

The 60% is measured in money, so it moves with import prices. The 101% assumes nothing is wasted.

So: Britain grows just enough, as long as gas, soya, lorries and weather all behave. What do we have to work with?

1. Defra, *Agriculture in the UK 2025*, ch. 14: the food chain
2. AIC soya report (~2.39 Mt of soya meal, 2024) and our arithmetic
3. Defra, *Agriculture in the UK 2025*, ch. 2: livestock use about 84% of farmland
4. Defra, *Nutritional supply of agriculture in the UK 2025*
5. Defra, *Farming evidence pack* (Sep 2026)
6. Defra, *Agriculture in the UK 2025*: food inflation peaked at 19.2% in March 2023

CHAPTER 2

Good land, but not much of it

> Britain is one of the best places on Earth to grow wheat and grass. There just isn't much land per person.

Britain is mild and wet. It rains all year, it rarely freezes hard, and there is no dry season.[1] That makes it superb for wheat and grass. A typical British wheat field grows about twice as much as the world average, and a farm in Lincolnshire holds the world record.[2,3]

But we are a crowded island. Each of us has about a third as much cropland as a person in France.[4] Much of our best land is in the east, which is getting drier, and in the Fens, where the dark peat soil is slowly wasting away. More than half of England's very best farmland lies where rivers or the sea can flood it.[5]

There is hope in the soil itself. In a long-running experiment at Rothamsted, fields rested under grass and clover for some years slowly gained back carbon, the stuff that makes soil rich and spongy.[6] Soil that has been worn out can recover, if we give it time.

2×

the world average: a typical British wheat harvest per hectare (~7.8 tonnes).[2]

~£1.2bn

a year: the cost of worn-out soil in England and Wales (2015 study).[7]

0.09 ha

of cropland per person in the UK, about three and a half tennis courts. France has ~0.28 ha.[4]

SOIL CAN HEAL



At Rothamsted, rested fields raised topsoil carbon from **0.98% to 1.23%** over 28 years, then levelled off. Fields cropped every year lost carbon.[6]

So: the land is good and the soil can heal, but there is little of it per person, and the weather it depends on is changing.

1. Met Office, UK climate zones
2. AHDB, final harvest report (2026); UK average derived from the survey
3. Guinness World Records, highest wheat yield (17.95 t/ha, 2022)
4. FAO arable land totals ÷ population (our arithmetic); a doubles tennis court is 261 m² (our comparison)
5. Environment Agency flood-risk mapping (2026): 59.2% of England's Grade 1 land lies in mapped river or sea flood-risk areas
6. Johnston et al. 2017, Rothamsted ley-arable experiment
7. Graves et al. 2015, *Ecological Economics* 119

CHAPTER 3

The weather is changing

> One bad week rarely hurts a harvest. A whole season of bad weather does, and that is happening more often.

British wheat is tough. It can shrug off a cold snap or a wet week. What hurts is when the bad weather piles up across a whole season: an autumn too wet to sow, a spring too dry for the seed to grow, a summer too wet to harvest.[1]

That pile-up is becoming normal. Four of England's five worst harvests in the last forty years came between 2020 and 2026.[2] 2025 was the warmest year ever recorded in the UK.[3] New crop diseases and animal diseases are moving north as winters get warmer.[4,5]

And when Britain has a bad year, our neighbours often do too. In 2026 drought hit Britain, France and Germany at the same time.[6] That matters, because the place we would normally buy from is having the same bad year. Trade cannot rescue us from a failure that everyone shares. A store of food can.

4 of 5
of England's worst harvests since 1984 came in 2020–2026.[2]

~8 Mt
of grain lost in France alone in the 2026 drought.[6]

−11%
wheat harvest per field in 2026, compared with the last five years.[7]

READ WITH CARE
Average harvests may hold up for decades; the danger is in the bad years. Some of 2026's losses came from farmers planting less because fertiliser was so expensive, not from weather alone.

1.2%
a year: the chance that the world's main wheat regions all fail together. It used to be 0.3%.[8]

So: bad years are coming more often and hitting our neighbours at the same time. And a harvest needs much more than weather.

1. Slater et al. 2022, *Earth System Dynamics* 13
2. ECIU, UK harvest even worse than previously thought (Oct 2026) (four crops, by volume)
3. Met Office (2026)
4. EPPO, wheat stem rust in the UK and Ireland
5. GOV.UK, bluetongue
6. Carbon Brief, Europe's 2026 heat and drought in six charts
7. AHDB, final harvest report (2026)
8. Gaupp et al. 2020, *Nature Climate Change*

CHAPTER 4

What a loaf really needs

A loaf of bread runs on gas, diesel, ships, electricity and computers.

Follow one loaf back to the field. The wheat grew because a farmer spread nitrogen fertiliser, and that fertiliser is made from natural gas. Britain no longer makes it in any quantity: its last big fertiliser factory, at Billingham, closed in 2023.[1] All of Britain's phosphate, the other key plant food, is imported too.[2]

The grain went by lorry, and modern lorries need a fluid called AdBlue, made from the same stuff as fertiliser. Without it, the lorry slows to the speed of a bicycle.[3] The flour was milled with electricity. The loaf was ordered by computer. And much of the rest of our food has to be kept cold: more than half of what we eat is chilled or frozen.[4]

Each of these links has already snapped somewhere. In 2025 a computer attack on one meat supplier stopped deliveries to most big supermarkets.[5] That same spring, a blackout across Spain and Portugal stopped card payments, warehouses and fridges for 50–60 million people.[6]

ONE SHORT SEA CROSSING

About **42%** of our imported dairy and eggs and **39%** of imported meat come across the short stretch of sea between France and Kent.[7]



>50%

of the food we eat must be kept chilled or frozen. No power, no cold.[4]

20 km/h

the top speed of a modern lorry once its AdBlue runs out.[3]

IMPORTANT, BUT NOT MAPPED

Food is officially one of 13 "critical" national systems. But the government's spending watchdog says nobody has yet listed the mills, grain dryers and cold stores it depends on.[8]

So: the weak links are energy, fertiliser and transport, not land. The biggest single one is food for our animals.

1. Defra, *UK Food Security Report 2024*, theme 3
2. Lancaster University, UK Phosphorus Transformation Strategy (2022)
3. Reg. (EU) 582/2011, Annex XIII (retained in UK law)
4. Cold Chain Federation, policy strategy 2026
5. Food Ingredients First, ransomware at a UK supermarket supplier (2025)
6. Slimstock, the Iberian blackout and the supply chain
7. Defra, *UK Food Security Report 2024*, theme 3 (the "Short Straits": Dover and Folkestone to Calais and Dunkirk)
8. National Audit Office, *Resilience of the food supply chain to disruptions* (Sep 2026)

CHAPTER 5

Grain, animals and the sea

> Our grain is not the weak point. The soya we import for animals is. And the grain we feed to animals is a hidden reserve.

Here is a surprise: about half of the wheat, barley and oats grown in Britain is fed to animals, not people.[1] That is not wasted, but it is costly. It takes several plates of grain to make one plate of meat from a grain-fed animal.[2]

In a crisis, that animal grain becomes a hidden reserve. Feed barley can become porridge, flatbread and wholegrain food, though not quite today's soft white loaf.[3] Our model shows that if imports stopped in a normal year, British grain could keep both people and animals going, as long as we stopped turning grain into whisky and car fuel.[4]

The real gap is protein. Pigs and chickens rely on imported soya. Cows and sheep do best on grass, on hills and wet land where crops cannot grow.[5] And around our coasts swim mackerel and herring we mostly sell abroad.[6]

IF IMPORTS STOPPED FOR THREE YEARS (OUR GRAIN MODEL) [4]

What the harvests are like	Extra grain we would need
Normal harvests, animals fed as now	2.3 million tonnes
Two harvests 15% down, animals get 80% of their feed	2.5 million tonnes
A fertiliser shortage cuts one harvest by a third	3.5 million tonnes
Three harvests 30% down	11.4 million tonnes: too much to store. We would also need fewer animals

~52%

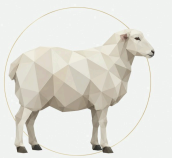
of UK wheat, barley and oats goes to feed animals.[1]

GROWING OUR OWN PROTEIN

Replacing all imported soya would take field beans on about a sixth of our cropland. A third within ten years is realistic.[4]

ANIMALS ON THE RIGHT LAND

Keep cattle and sheep on grass that nothing else can use. Match pig and chicken numbers to the protein we can grow at home.[5]



THE SEA

40%

of the fish we eat is now caught or farmed in the UK; it was 89% in 1975.[6]

So: grain is a safety buffer if we plan ahead; imported soya is the gap. Both depend on a world that is getting more dangerous.

1. Defra cereal usage statistics (2023 data)
2. Our arithmetic: 3 kg of grain ≈ 9,900 calories, against ~1,500–2,500 in 1 kg of meat; for world context see Mottet et al. 2017, *Global Food Security*
3. UK Flour Millers, annual review
4. Our grain model, `cereal_balance.py`: imports stop on 1 July; distilling and bio-fuel cut by 75%
5. Wilkinson 2011, *Animal* 5(7): only milk and upland suckler beef returned more human-edible protein than they ate
6. Harrison et al. 2023, *Reviews in Fish Biology and Fisheries*

CHAPTER 6

When the world gets dangerous

> In 2026 two of the world's key sea routes were disrupted. Britain buys little directly through either, and still pays for both.

This year, the Strait of Hormuz, the narrow sea passage out of the Persian Gulf, was closed.[1] Up to 30% of the world's traded fertiliser passes that way. At the same time, ships carrying grain on the Black Sea came under attack.[2] Within weeks, the price of a common nitrogen fertiliser more than doubled.[3] World food prices are now the highest in almost four years.[4]

So far, British shoppers have barely noticed: food prices rose only 1.3% in the year to September.[5] But fertiliser bought dear this winter shows up in next year's harvest and next year's prices.

History tells us what happens when food gets scarce. In 2022, 32 countries banned or limited food exports.[6] When food is short, money matters less than having it already. Countries that hold stock do not have to panic-buy, and they can help their friends.

up to 30%

of the world's traded fertiliser passes through the Strait of Hormuz.[1]

$400 → $850

per tonne of urea fertiliser, from early 2026 to April.[3]

77

food export bans or limits, by 32 countries, in 2022.[6]

A RESERVE IS ALSO FOREIGN POLICY

A country with food in store can keep its own exports open, help partners, and in calm years sell old stock to aid agencies at prices agreed in advance.

READ WITH CARE

Contracts to buy food later can fail when countries ban exports, so food at home comes first. We could not confirm whether Hormuz was open in late October 2026.

So: when food is short, stock beats money. And when food gets scarce or dear, people notice who is treated fairly.

1. FAO, chief economist on disruption to the Strait of Hormuz (2026)
2. S&P Global, Black Sea grain trade factbox (Aug 2026)
3. WTO data blog (10 Jul 2026)
4. Fastmarkets, FAO Food Price Index, September 2026 (136.0, highest since Nov 2022)
5. ONS, consumer price inflation, September 2026
6. IFPRI, food export restrictions tracker

CHAPTER 7

When food runs short, fairness matters

People forgive shortage sooner than unfairness.

When food gets dearer, people get angrier. Studies across many countries find that rising food prices are followed by more unrest.[1] In a rich country like Britain, the first thing to fail is not the harvest but the delivery: in the 2000 fuel blockade, supermarkets warned they would run out within days.[2]

What turns a shortage into chaos is the feeling that it is unfair. In Egypt in 1977, cutting the bread subsidy caused riots in almost every big city; it was reversed in two days.[3] In Japan in 2025, rice prices nearly doubled, the emergency rice reached shops too slowly, and the farm minister resigned.[4] In Britain during the Second World War, rationing gave everyone the same basic share, and baby deaths actually fell.[5]

Many families are already on the edge. One household in eight cannot always afford enough food.[6] Only one in twenty-five has three days of water at home.[7]

12.2%
of UK households struggle to get enough food (July 2026).[6]

2.6 m
emergency food parcels given out by food banks in 2025.[8]

BRITAIN IS NOT READY

The government's spending watchdog found local plans for food emergencies "insufficient", and no power for the government to direct food firms in a disaster.[9] In October 2026 more than 150 experts wrote to the Prime Minister asking for national food stores.[10]

READ WITH CARE

In Britain the likely problem is panic buying, empty shelves and lost trust, not riots. Unrest has many causes; prices are one.

So: being prepared, openly and fairly, keeps the peace. Other countries already do this.

1. Bellemare 2015, *American Journal of Agricultural Economics* 97(1)
2. CNN transcript, UK fuel protests (13 Sep 2000)
3. NPR, protests and the price of bread (2011)
4. Food Manufacturing, Japan's rice crisis (2025)
5. Working paper citing Ministry of Health data (arXiv 2301.09982)
6. Food Foundation, food insecurity tracker (Jul 2026)
7. UK public survey of risk perception and preparedness 2026 (4% have three days of water)
8. Trussell, food banks provide over 2.6 million parcels
9. National Audit Office, summary (Sep 2026)
10. Global Government Forum, experts' letter (Oct 2026)

CHAPTER 8

What other countries do

Many countries like ours keep food in reserve. Britain keeps none.

Finland, a country that remembers hunger and war, keeps at least six months of bread grain.[1] Switzerland requires its food firms to hold two to four months of staples on top of their normal stock.[2] Norway is rebuilding a grain store it let go.[3] Sweden passed a new law this year.[4]

None of this is exotic. It is ordinary, boring, careful government. And it works: in August 2026, when low rivers stopped barges reaching Switzerland, the Swiss simply released animal feed from their stores.[5] The lesson from Japan is the opposite one: having food is not enough. You also need clear, fast rules for letting it out.[6]

Country	What it keeps	How
Finland	At least 6 months of bread grain[1]	A state agency, paid for by a small fee
Switzerland	2–4 months of wheat, sugar, rice, oil, feed and fertiliser[2]	Food firms must hold extra stock by law
Norway	~3 months of bread grain by 2029[3]	State grain, stored by millers
Sweden	Grain, seed, fertiliser, feed and spare parts[4]	New law from 1 July 2026
Germany	About 1 month of bread grain, plus rice and pulses[7]	State reserve in ~150 private stores
Japan	Rice reserve, target 1 million tonnes[6]	Bought by the state, kept ~5 years
Singapore	Two months of rice[8]	Required of every rice importer
UK	**None**[9]	

⚠ READ WITH CARE

Finland's and Switzerland's costs also cover fuel and medicines, so they are not like-for-like prices for a UK food reserve.[10] Much national reserve data is incomplete or secret.

So: keeping food in reserve is normal in countries like ours. Here is what a British one would hold and cost.

1. NESA (Finland), additional grain for emergency stockpiles
2. réservesuisse, compulsory stocks (1 Jan 2026)
3. Norwegian government, emergency grain storage
4. Riksdag, committee report on food stockpiles (SFS 2026:837)
5. cash.ch, Swiss protein-feed release (Aug 2026)
6. Asia News Network (2025); Adnkronos (Sep 2026)
7. top agrar, Germany's grain reserves
8. MTI Singapore, reply on stockpiling
9. UK Parliament, written answer HL15829
10. NESA, funding; SRF, the Swiss pantry (2019) (~CHF 14 per person)

STORE: a national food cupboard

> About £10 per person a year, under 3p a day. Bread for everyone, a no-cook food pack for everyone, and the seed for next year. Nobody left out.

Imagine the power has been off for three days. Ovens are cold, card machines are dead, the shops are emptying. What do people actually need? We built the reserve backwards from that moment, starting with the meal.

First, bread. Four million tonnes of good milling wheat, kept by millers and grain merchants on top of their normal stock, would keep the bakeries going for about ten months even with no imports and no harvest, as long as the mills have power and staff.[1,2] **Second, food that needs no cooking**: a three-day pack of crispbread, biscuits and oatcakes for every single person, stored near where people live, plus vitamin tablets for 90 days. **Third, the people who need something different**: baby milk, medical foods, soft food. **Fourth, the next harvest**: seed, fertiliser stored safely, animal feed, and months of rice, oil and sugar.[1]

ALWAYS FRESH

About a fifth of the stock is used and replaced each year, so it never goes stale. Old stock goes to schools, the armed forces and food banks.



Think of it as the National Insurance (not National Insurance).

Prof. Gen. Adm. F.L. Amingo, Battalion of Militarized Satire

WHO HOLDS IT

Part is owned by the state and paid from taxes. The rest is extra stock that food firms must hold by law, as in Switzerland, with the cost added to food prices.[3]

~10 days

of food in the shops today. Wheat stocks before harvest fell to about 39 days of use.[4,5]

IF MONEY IS SHORT, BUILD FROM THE BOTTOM UP

Total cost per person a year at each step.[1]

Up to	What it adds
£2.71	**Bread for everyone:** 4 million tonnes of milling wheat, strong wheat, vitamins for flour, sacks
£3.48	+ **no-cook food for everyone:** 3-day pack, vitamin tablets, bottled water
£6.92	+ running it: a small agency, local plans, fuel and backup power, delivery to the door
£7.55	+ babies and people with special diets
£9.19	+ next year's harvest: seed, fertiliser, animal feed
£10.33	+ imported basics: rice, oil, sugar, beans, oats, milk powder, tinned fish

So: the food cupboard starts with the meal and covers everyone. Here are the numbers.

1. Our cost model, `reserve_basket.py`
2. UK Flour Millers, annual review (4.8 Mt milled, 2025)
3. réservesuisse, compulsory stocks
4. Full Fact, the UK's food supply
5. AHDB, end-of-season UK cereal supply and demand (Sep 2026) (1.514 Mt)

The food cupboard in numbers

> £718m a year, about 6.8 million tonnes of food, and an honest answer to everything that could go wrong.

What	How much	£ per person a year
Milling wheat, 4 million tonnes (half in new stores), plus strong wheat for bread	4.35 Mt	2.62
No-cook food pack for everyone, 3 days (~500 g a day)	104 kt	0.40
Vitamin tablets, flour vitamins, sacks, 90 million litres of bottled water	–	0.46
Running it: a small agency and inspectors (~£33m), local plans, fuel and power, delivery, security	–	3.44
Baby milk (~7,000 t), medical foods, special no-cook food for 5 million people	–	0.63
Next harvest: seed, safely stored nitrogen fertiliser, 10 weeks of soya feed	1.21 Mt	1.64
Imported basics: rice and oil (~4 months), sugar (~2 months), beans, oats, milk powder, tinned fish	1.18 Mt	1.14
Total	**~6.8 Mt**	**~£10.33 (£718m)**

Source: our cost model.[1]

£4.93 · £5.40
paid through taxes · through food prices.[1]

~£3.7bn
of food and stores to pay for up front, by the state and firms, over ~5 years.[1]

If things cost more:[1] cheapest case £9.40 · our central case **£10.33** · prices 30% higher £11.35 · firms borrow at 8% £11.92, at 10% £12.73 · grain and fertiliser prices double £13.73 · bad case £16.56 · very bad case £18.34.

⚠ WHAT COULD GO WRONG, AND WHAT WE DO ABOUT IT

Worry	Answer
It adds to government debt	Only the ~£0.72bn public part is meant to count. If the statisticians count the firms' stock too, debt rises by ~£2.2bn more. Ask them first
Firms pretend normal stock is reserve	The duty is on top of a registered normal level, checked by independent inspectors (£8m a year)
The first week fails without water and delivery	90 million litres covers 10 million people for 3 days; water firms must cover the rest and practise it. A national food drop would cost ~£400m more
The same wheat is counted twice	The 4 million tonnes is for bread only; feeding animals through two bad harvests needs ~2.5 million tonnes more[2]
Stored fertiliser explodes	We never store the explosive kind (ammonium nitrate, the Beirut chemical), only safer forms[3]
Refilling is costly	Refill within three years, when prices and quality are right; refilling 30% dearer costs ~£0.25bn
It gets raided to fight inflation	Released only on published supply triggers, never to manage prices

So: a store buys time, not independence. What we grow with that time is the second pillar.

1. Our cost model, `reserve_basket.py` (central case and sensitivities)
2. Our grain model, `cereal_balance.py`
3. HSE, ammonium nitrate and COMAH

GROW: healthy soil, water and seed

> A store buys time. Soil, water and seed decide what we can grow with it.

A food cupboard gets us through a bad season. It cannot fix the deeper problem: every British harvest leans on imports and luck. Our farms used about 904,000 tonnes of nitrogen fertiliser last year, made from gas, and Britain no longer makes the key ingredient itself (chapter 4).[1]

So the second pillar looks after the things every harvest stands on. **Soil**: test it, measure how wet it is, and help it recover. **Water**: catch winter rain in farm reservoirs for dry summers. **Seed**: breed and multiply our own. **Nutrients**: use manure and waste wisely, and grow more beans and peas, which make their own nitrogen. Today about 50 stations measure soil moisture across the whole UK.[2] We propose ten times as many.

Area	What the money buys	£m a year
Soil and water checks	~500 soil-moisture stations (from ~50), satellite checks, a national soil-health map, regular field tests	35
Water storage	Farm and local reservoirs that fill in winter	100
Seed	Seed banks, growing our own seed (including beans and peas), public plant breeding	40
Nutrients	Testing and careful use of manure; recovering phosphate; beans and peas; a pilot for making fertiliser with green power	60
Switching to soil-building farming	Independent advice, farmer groups, payments to bridge the first lean years	50
Home-grown protein	Mills and plants to process beans and peas	25

Source: our cost model.[3]

~£310m

a year of new money: **~£4.46 per person**.[3]

SMALL TODAY

Reservoir grants have been £10–15m a round.[4] The UK's vegetable seed bank holds about **14,000** samples.[5]



BETTER USE OF MONEY WE ALREADY SPEND

Britain already spends about **£4.87bn** a year on farm support, about £70 per person.[6] We ask that it buy a safer food supply. Some farms will lose out, and that should be said openly.

READ WITH CARE

£310m is a start and a set of trials, not full independence. Soil-building farming often makes more profit after 3–5 years, but paying back the switch takes about nine without help.[7] Pay for measured results, not labels.

So: grow money goes where we depend on others most. And it pays back beyond food.

1. Defra, British Survey of Fertiliser Practice 2025
2. COSMOS-UK soil moisture network, *ESSD* (2021)
3. Our cost model, `reserve_basket.py`
4. Defra Farming blog, new drought support (Aug 2026)
5. Farming UK, Warwick seed bank
6. Defra, *Agriculture in the UK 2025*, ch. 10–11
7. WBCSD, *Closing the gap* (regenerative agriculture in Europe)

Healthy soil pays back twice

> Farming that rebuilds soil is said to do many good things. Some are proven, some depend on how it is done, some are about values. We say which is which.



PROVEN

Healthier soil at the places studied; less soil washed away; more wildlife on hedges and field edges; better mood from outdoor work.[1,2]

WEAK OR LIMITED

More nutritious food: weak or mixed evidence. Carbon stored in soil: real but limited; it rises, then levels off.[1]

DEPENDS HOW IT IS DONE

Water held in the land; lower costs; coping with drought; cleaner air (only if less fertiliser is used); local jobs (more work per field, but no proof of more jobs overall).[3]

VALUES, NOT MEASUREMENTS

Learning, community, a sense of connection, and a duty to the people who come after us.

So: we promise only what the evidence supports. Next: who looks after all this.

1. Johnston et al. 2017, Rothamsted ley-arable experiment
2. Coventry et al. 2021, nature-based outdoor activity and mental health
3. Organic farming and employment, review

CHAPTER 11

Who looks after it

> A small, open team, helped by machines. You cannot hide forty warehouses, so we do not try.

A food reserve does not need a giant ministry. We propose a **small agency of about 65 people**, never more than 100 even in a crisis. Every job has a lead and two trained back-ups, so illness or holidays never stop anything.[1] Independent inspectors check the stores.

Machines do the watching. Sensors in every store check temperature and damp; computers track stock, prices, weather and shipping day and night.

People make the decisions, check the work, and answer to Parliament. Two people must sign before any public stock is released.

And everything is public by default: where the stores are, how much is in them each month, what it costs, and what went wrong. Anyone who finds a real weakness gets a reward. Only a short, published list stays secret, such as the exact sites of fuel, baby milk and fertiliser.

NEAR PORTS, NEAR PEOPLE



8–10 port stores at grain ports such as Immingham, Liverpool and Avonmouth. **About 30 district stores** across all four nations, away from flood zones. **Thousands of local points** (schools, council depots) holding the no-cook food pack.

~£33m

a year for the agency, sensors, data and inspectors.[1]

~1,070

jobs across stores, labs, haulage and local planning.[1]

READ WITH CARE

65 is a starting team, to be resized once the real work is known. Computers can fail first in a blackout, so everything must also work on paper and radio, and be practised that way.

So: a small, visible team runs a big, boring network. Both pillars also create work worth having.

1. Our cost model and staffing plan, `reserve_basket.py`; research note 23

CHAPTER 12

Good work on the land

> Make growing food a job people are proud of, let machines do the drudgery, and grow enough to sell in good years.

Two things are happening at once. Computers and AI are taking over many first jobs in offices: graduate job adverts are at their lowest since 2016.[1] Meanwhile our farmers are getting older, and few young people are following them.

Both pillars need people: in stores and labs, and as soil testers, water engineers, seed breeders and bean processors. But today farm work is often lonely, seasonal and stressful. Nine in ten young farmers say poor mental health is the industry's biggest hidden problem.[2] Slogans will not fix that. Pay, training, housing and teamwork will.

There is good evidence that working outdoors helps. A review of studies found that structured outdoor activity clearly reduced depression and anxiety.[3] In the 1930s, young Americans who worked in the Civilian Conservation Corps lived longer and earned more over their lives.[4]

up to ~3,600

jobs across both pillars (before counting jobs lost elsewhere).[5]

5–11%

of fruit-picking jobs were filled by UK residents in the 2020 "Pick for Britain" campaign.[6] Adverts alone do not move people.

A FOOD-GROWING NATION

£25.6bn

of food and drink exported in 2025, against £66.9bn imported.[7] We could sell more of what we grow best: oats, barley, dairy, lamb, seed potatoes.

⚠ READ WITH CARE

No study yet shows people who lose office jobs to AI moving into land work. Pride follows pay, land and housing, not slogans.

So: the people matter as much as the grain. Paying for all of it fairly needs honest prices.

1. Adzuna data via VnExpress (Jul 2026)
2. Farmers Guardian, Farm Safety Foundation survey (2025) (91%)
3. Coventry et al. 2021, *SSM – Population Health* (standardised effects −0.64 depression, −0.94 anxiety)
4. NBER working paper w27103
5. Our jobs estimate, research note 25
6. FarmingUK, NFU harvest 2020 survey
7. FDF, trade snapshot 2025

CHAPTER 13

Paying the true price of food

> Charge for what does harm, pay for what protects, and give the money back to people.

Cheap food has costs we do not see at the till: polluted rivers, worn-out soil, illness from poor diets, and the gases that heat the planet. Farming makes about an eighth of Britain's greenhouse gases.[1] Worldwide, these hidden costs add up to at least $10 trillion a year.[2]

Our position is simple: **polluters pay, and people who clean up get paid.** One fair carbon price for everyone, with no free passes. The money goes back to farms and families, so nobody is made poorer for doing the right thing. Britain is already starting: from January 2027, imported fertiliser will carry a carbon charge at the border.[3]

Denmark shows how to do it with farmers rather than against them. It will charge for livestock emissions from 2030, but give most of it back to farming.[4] Europe's 2024 farmer protests showed the other way: change forced on farmers fails.[5]

~12%

of UK greenhouse gases come from farming (2024).[1]

£0.4–1bn

a year: Danish-style livestock charges applied in the UK, all of it given back (our estimate).[4]

⚠ **BEWARE WEAK PROMISES**

Payments for storing carbon in soil have been shown to follow rainfall more than farming in Australia.[6] Pay for what is measured, not what is claimed.

⚠ **READ WITH CARE**

Fair carbon prices raise food prices, so the money must go back to people. Charging British farms but not imports just moves the pollution abroad.

So: honest prices pay for the change. Here is the plan, step by step.

1. Defra, *Agriculture in the UK 2025*, ch. 10–11
2. FAO, *The State of Food and Agriculture 2023*
3. HM Treasury, carbon border adjustment factsheet
4. The Local Denmark, livestock CO_2 tax (2024); UK range: our estimate
5. Euronews, EU withdraws pesticide law (Feb 2024)
6. Mitchell et al. 2024, *Carbon Management*

CHAPTER 14

The plan, step by step

> Store and grow. Write the crisis rules now, in calm, not later, in panic.

WHEN

When	Store	Grow
First year	Pass the law, set a minimum reserve and release rules; check the debt question; buy grain only in good harvests; local food plans everywhere	Start soil testing and moisture stations; more reservoir grants; aim farm support at soil, water and protein
Years 1–3	Firms start holding extra stock; build the public reserve and the no-cook food pack; agree and practise the crisis rules	Grow our own seed; first reservoirs filling; manure testing; bean and pea mills
Years 3–10	A full, rotating, audited reserve in all four nations	A third less imported soya; a national soil map; a green-fertiliser pilot

WHAT HAPPENS IN A CRISIS, LEVEL BY LEVEL

Level	What is going on	What we do
0 · Normal	Everyday life	Stock is used and replaced; monthly public report; practice drills
1 · Disruption	Days to weeks: fuel blockade, cyber attack, power cut	No-cook food; fuel and power for mills and depots first; limits on how much one shopper can buy
2 · Long shortage	Months: imports cut, fertiliser shortage, one failed harvest	Release stock in stages; animal grain to people; less grain to whisky and fuel; fewer animals bred
3 · Severe	Two bad harvests plus a fertiliser shortage	**Fair Shares Guarantee**: the same basic share for everyone, extra for those who need it, capped prices on basics, paper backup, a set end date
4 · Catastrophe	Big volcano, war, a global crop disease	All grain we can eat goes to people; help from and to other countries

FEWER ANIMALS, PLANNED WITH CARE

Written in advance with farmers and vets: stop new breeding first, keep processing simple, pay farmers fairly, and never let animals starve.

THE LAW

Today's emergency powers last only 30 days at a time.[1] A Food Security Act should set up the reserve for the long term. The Cabinet Office decides when to release; Defra runs it, with all four UK governments involved.

So: store and grow, run by rules written in calm. One related idea sits outside the costed plan.

1. Civil Contingencies Act 2004, Part 2 (ss. 26–27): emergency regulations must be approved within 7 days and lapse after 30

A RELATED IDEA · NOT PART OF THE COSTED PLAN

Homes, land and a living

> Build homes where the work is. Give a new generation a home, land to work and a reason to stay.

Britain has built whole new places before: the garden cities of Letchworth and Welwyn, the post-war new towns, Milton Keynes, Cambourne near Cambridge,[1] and now Salt Cross garden village in Oxfordshire, planned for about 2,200 homes.[2] Since COVID, many people no longer have to live near an office.

The idea: build modern, warm, cheap-to-run homes in a cluster next to an existing village, with **shared** farmland around them. The land is owned by a community trust; tractors, a packing shed and a cold store are shared; residents can work in food and nature for some or all of the week. Not a plot each, but shared tools, skills and money to start.

GROUND RULES



Your home never depends on the farm job. Homes stay cheap through who owns them, not through labels. Keep the best land for food; never build on peat or flood plains. Test two or three sites first.

~9.6%

of UK farmland would go if a million homes each got four acres. That is why the land must be shared.[3]

~£500k

of sales a shared farm needs to pay four people; the village alone buys about £52k.[3]

READ WITH CARE

More homes help prices locally, but national house prices depend mostly on interest rates.[4] Planning is slow and contested.

So: a pilot worth testing, not a promise. The last page lists what still needs checking.

1. South Cambridgeshire District Council, new communities
2. West Oxfordshire DC, Salt Cross plan adopted (Feb 2026); Salt Cross FAQs
3. Our land arithmetic, research note 26
4. Miles & Monro 2019, Bank of England staff working paper

OPEN TO EVERYONE

Tell us where this breaks

> We publish our own doubts. Strong plans get stronger when people try to break them.

Every number in this briefing has a source you can click, and every source has been through four rounds of checking by reviewers whose job was to find mistakes. In the full evidence file each claim carries a label: from an official source, from a secondary source, our own estimate, our opinion, or not yet checked.

Some questions can only be answered by doing. Before anyone spends real money, these need testing.

⚠ STILL TO CHECK

1. Whether official statisticians would count the firms' stock as government debt, and what firms would really pay to borrow for it.
2. Whether each mill can switch to emergency flour mixes, with backup power.
3. A no-cook food pack designed by dietitians, with versions for babies, pregnancy, coeliac disease, allergies and people who cannot chew.
4. Door-to-door delivery and water, costed and tested in a real local drill.
5. Real quotes for storage, new stores, reservoirs and fuel.
6. Full UK 2026 harvest figures (due December 2026) and the government's reply to our information request.[1]

🔗 EVERYTHING IS OPEN

- The full evidence file: every claim, source and counter-argument.[2]
- Two models: the cost of both pillars, and month-by-month grain stocks.[3]
- 27 research notes.[4]

An army defends the border. A food reserve defends the dinner table.

1. WhatDoTheyKnow, FOI "Strategic food reserves and compound risk" (Oct 2026)
2. Full evidence file, `draft/briefing_full_context.md`
3. Models, `research/model/`
4. Research notes, `research/notes/`